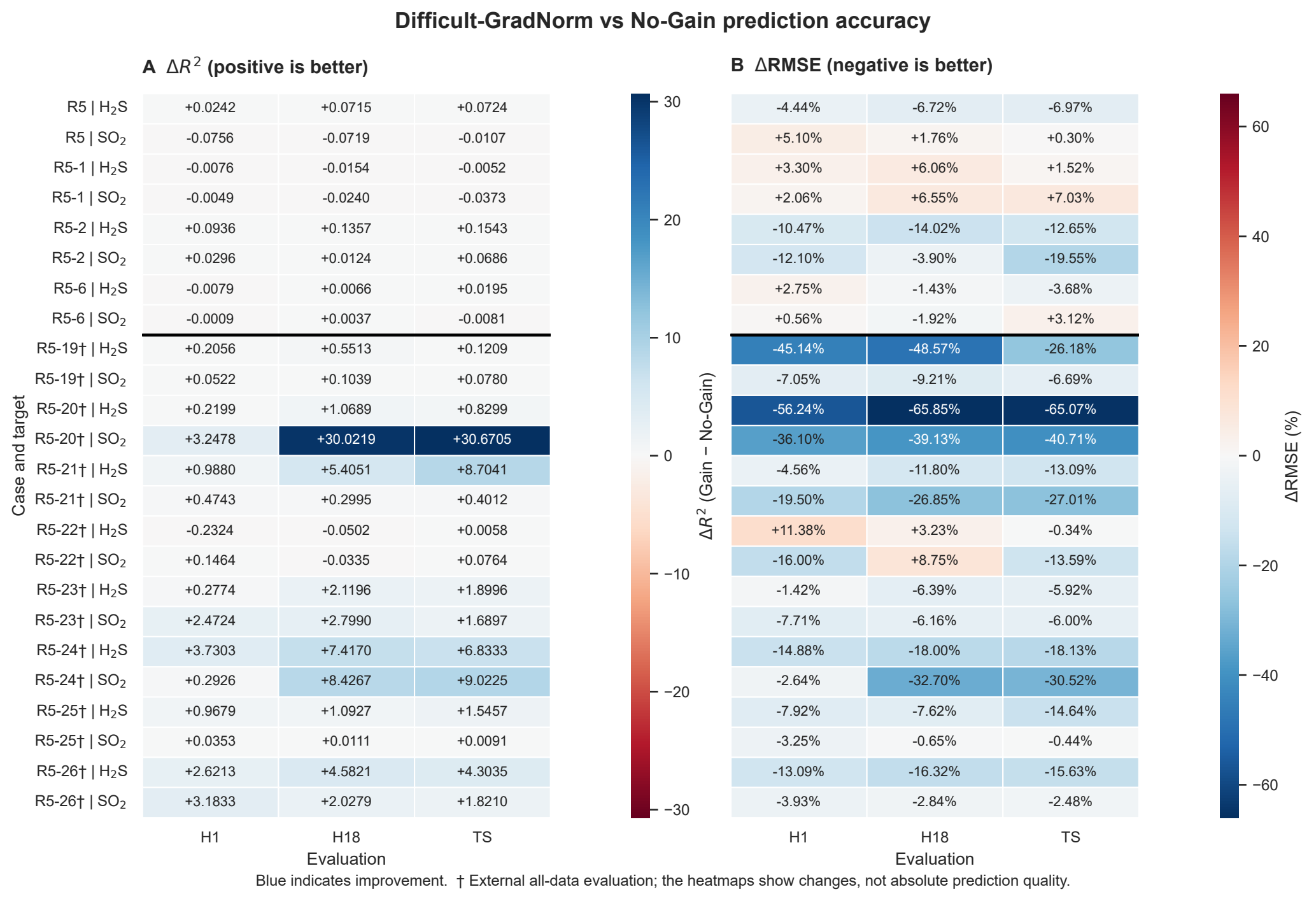

Difficult-GradNorm vs No-Gain prediction accuracy
A ΔR² (positive is better)
B ΔRMSE (negative is better)
Case and target
R5 | H2S
R5 | SO2
R5-1 | H2S
R5-1 | SO2
R5-2 | H2S
R5-2 | SO2
R5-6 | H2S
R5-6 | SO2
R5-19† | H2S
R5-19† | SO2
R5-20† | H2S
R5-20† | SO2
R5-21† | H2S
R5-21† | SO2
R5-22† | H2S
R5-22† | SO2
R5-23† | H2S
R5-23† | SO2
R5-24† | H2S
R5-24† | SO2
R5-25† | H2S
R5-25† | SO2
R5-26† | H2S
R5-26† | SO2
+0.0242 +0.0715 +0.0724
-0.0756 -0.0719 -0.0107
-0.0076 -0.0154 -0.0052
-0.0049 -0.0240 -0.0373
+0.0936 +0.1357 +0.1543
+0.0296 +0.0124 +0.0686
-0.0079 +0.0066 +0.0195
-0.0009 +0.0037 -0.0081
+0.2056 +0.5513 +0.1209
+0.0522 +0.1039 +0.0780
+0.2199 +1.0689 +0.8299
+3.2478 +30.0219 +30.6705
+0.9880 +5.4051 +8.7041
+0.4743 +0.2995 +0.4012
-0.2324 -0.0502 +0.0058
+0.1464 -0.0335 +0.0764
+0.2774 +2.1196 +1.8996
+2.4724 +2.7990 +1.6897
+3.7303 +7.4170 +6.8333
+0.2926 +8.4267 +9.0225
+0.9679 +1.0927 +1.5457
+0.0353 +0.0111 +0.0091
+2.6213 +4.5821 +4.3035
+3.1833 +2.0279 +1.8210
H1
H18
TS
Evaluation
ΔR² (Gain − No-Gain)
30
20
10
0
−10
−20
−30
-4.44% -6.72% -6.97%
+5.10% +1.76% +0.30%
+3.30% +6.06% +1.52%
+2.06% +6.55% +7.03%
-10.47% -14.02% -12.65%
-12.10% -3.90% -19.55%
+2.75% -1.43% -3.68%
+0.56% -1.92% +3.12%
-45.14% -48.57% -26.18%
-7.05% -9.21% -6.69%
-56.24% -65.85% -65.07%
-36.10% -39.13% -40.71%
-4.56% -11.80% -13.09%
-19.50% -26.85% -27.01%
+11.38% +3.23% -0.34%
-16.00% +8.75% -13.59%
-1.42% -6.39% -5.92%
-7.71% -6.16% -6.00%
-14.88% -18.00% -18.13%
-2.64% -32.70% -30.52%
-7.92% -7.62% -14.64%
-3.25% -0.65% -0.44%
-13.09% -16.32% -15.63%
-3.93% -2.84% -2.48%
H1
H18
TS
Evaluation
ΔRMSE (%)
60
40
20
0
−20
−40
−60
Blue indicates improvement. † External all-data evaluation; the heatmaps show changes, not absolute prediction quality.